Prove that the curl of a gradient is always zero.

Solution:

For a function T, show

$$\nabla \times (\nabla T) = 0$$

Using the definitions of the gradient and curl

$$\nabla T = \frac{\partial T}{\partial x}\hat{\boldsymbol{x}} + \frac{\partial T}{\partial y}\hat{\boldsymbol{y}} + \frac{\partial T}{\partial z}\hat{\boldsymbol{z}}$$

$$\nabla \times \boldsymbol{v} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right)\hat{\boldsymbol{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right)\hat{\boldsymbol{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)\hat{\boldsymbol{z}}$$

results in

$$\nabla \times (\nabla T) = \underbrace{\left(\frac{\partial^2 T}{\partial y \partial z} - \frac{\partial^2 T}{\partial z \partial y}\right)}_{=0}\hat{\boldsymbol{x}} + \underbrace{\left(\frac{\partial^2 T}{\partial z \partial x} - \frac{\partial^2 T}{\partial x \partial z}\right)}_{=0}\hat{\boldsymbol{y}} + \underbrace{\left(\frac{\partial^2 T}{\partial x \partial y} - \frac{\partial^2 T}{\partial y \partial x}\right)}_{=0}\hat{\boldsymbol{z}}$$

since the symmetry of second derivatives ensures that

$$\frac{\partial^2}{\partial x \partial y} = \frac{\partial^2}{\partial y \partial x}$$

$\square$